Prime63 in-flight admission engineering record

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This report records engineering observations only. It asserts no scientific experiment status, solver performance, relation yield, or end-to-end gain.

Scope

The successor preserves 27 frozen requests: six planted public controls and 21 natural systems across coset, random, and small-x arms at m=2 through 8. The unchanged guards are 300 seconds per cell, 8 GiB RSS, 8 GiB scratch, 64 models, and four hours per panel. The two prior launcher source hashes matched their archives.

The ordinary selector must say **ready** before dispatch. The standard supervisor creates the run directory and `launch.json` before the driver starts, changing the one-trial selector state to `needs_reconciliation`. The new admission path accepts that state only when the write-once launch record binds the exact plan SHA256, trial/run IDs, commands and a canonical absolute or repository-relative plan path, live owner/epoch claim, and fresh repository authorization. Parent and child each recheck these facts before additional task output or Sage work.

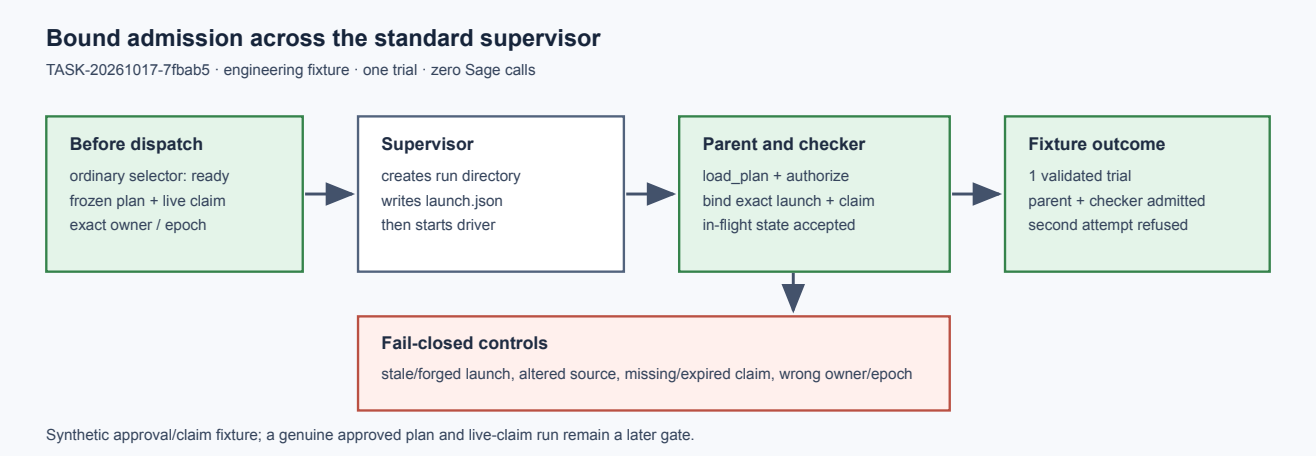


Figure 1: Observed fixture order and refusal boundary. Synthetic approval and claim do not establish admission for a future scientific trial.

Recorded engineering controls

Ten final tests passed. The one-trial fixture used the real standard supervisor, a disposable local Git repository, and real Python driver and checker subprocesses. It observed `ready` before dispatch and exact bound `needs_reconciliation` in both subprocesses. One fixture trial validated, the second attempt was refused, and zero Sage processes started.

Seven altered launch records were refused. A repository-relative plan token resolved to the exact frozen plan; a changed token was refused. Changed source, wrong owner/epoch, missing or expired claim, selector refusal, and prior receipt were also refused without additional task output. A small-prime exact checker accepted one polynomial/full-point root and rejected five altered objects. A raw archive check rejected changed bytes and passed replay with disk cell files removed. A bounded positive remained censored; a forged `ok` response without the planted witness was rejected. The 27-request schedule matched the archived launcher bytewise. The mock panel driver emitted a schema-valid manifest.

The first goal-lane publish attempt hit a sandbox Git index-lock denial; the same write-once claim was subsequently committed and pushed at `450209b3d2`. A repository `ResourceWarning` appears in final stderr; all ten tests passed. Exact commands, hashes, environment and raw logs are in this task directory.

Remaining gate

The Coordinator still needs an exact preapproval packet with `approvable: true`, an archived one-trial scientific plan that declares all fixed raw archive and receipt custody paths, and a genuine ready-selector/live-claim positive run. The checker can replay its raw cells from tar plus receipt without disk scratch files. No full-width system was solved in this engineering task. Degree of regularity, natural relation yield, useful rank, target work, matched rho, and end-to-end runtime remain unmeasured.